

Disproof of TLMC Conjecture 00000000325

The packing dimension of $W(\tau)$ is not $2/(1 + \tau)$

1 The conjecture

Conjecture 1 (TLMC 00000000325). *Let*

$$W(\tau) = \{\alpha : \liminf_{q \rightarrow \infty} q^{1/\tau} \|q\alpha\| = 0\}.$$

Then the packing dimension of $W(\tau)$ is exactly $2/(1 + \tau)$.

2 Verdict: FALSE

The conjecture is false. A single explicit counterexample suffices.

Claim 1 (Primary counterexample: $\tau = \frac{1}{2}$). *At $\tau = \frac{1}{2}$ the value asserted by Conjecture 1 is*

$$\frac{2}{1 + \tau} = \frac{2}{1 + \frac{1}{2}} = \frac{4}{3} > 1,$$

which exceeds the dimension of the ambient space $\mathbb{R}$.

Proof. Arithmetic: $2/(1 + 1/2) = 2/(3/2) = 4/3$, and $4 \cdot 1 = 4 > 3 = 1 \cdot 3$, so $4/3 > 1$. But the packing dimension is monotone under inclusion and $\dim_P(\mathbb{R}) = 1$; hence $\dim_P(A) \leq 1$ for *every* $A \subseteq \mathbb{R}$. No set can have packing dimension $4/3$, so the conjectured equality fails at $\tau = 1/2$. $\square$

Remark 1 (Second, independent reason at the same τ). Writing $v = 1/\tau$, the set $W(\tau)$ is the “exact order v ” set $\{\alpha : \forall \varepsilon > 0, \|q\alpha\| < \varepsilon q^{-v} \text{ i.m.}\}$. At $\tau = 1/2$ we have $v = 2$, and the Khintchine divergence test asks for the divergence of

$$\sum_q q \cdot \varepsilon q^{-v} = \varepsilon \sum_q q^{1-2} = \varepsilon \sum_q \frac{1}{q},$$

the harmonic series, which diverges. Hence $W(1/2)$ has *full Lebesgue measure*, so its packing dimension equals 1 — already different from the conjectured $4/3$ even ignoring the impossibility argument above.

3 Boundary of the attack

For rational $\tau = p/q > 0$ put $v = q/p$. The conjectured value is $2/(1 + \tau) = 2q/(p + q)$.

- $0 < \tau < 1$ (**i.e.** $q > p$): the conjectured value $2q/(p + q)$ exceeds 1, because $2q > p + q \iff q > p$. The conjecture is impossible on the entire range $0 < \tau < 1$ by the monotonicity argument; this includes the primary counterexample $\tau = 1/2$.

- $0 < \tau < 1/2$ (**i.e.** $v > 2$): here $W(\tau)$ is a null set and the Jarník–Besicovitch theory gives $\dim_H W(\tau) = \dim_P W(\tau) = 2/(v+1) = 2\tau/(1+\tau)$, which equals the conjectured $2/(1+\tau)$ only at $\tau = 1$. E.g. at $\tau = 1/4$ the true value is $2/5$ while the conjecture claims $8/5$.
- $\tau \geq 1/2$ (**i.e.** $v \leq 2$): Khintchine’s theorem gives full measure, so $\dim_P W(\tau) = 1$, while the conjectured $2/(1+\tau)$ differs from 1 at every $\tau \neq 1$; e.g. at $\tau = 2$ the conjecture claims $2/3$ but the true value is 1.
- $\tau = 1$ (**the only non-false point**): the conjectured value is 1, matching the full-measure value.

Thus the conjecture fails for every $\tau > 0$ except the trivial point $\tau = 1$, and it is outright impossible (value exceeding the ambient dimension) for all $0 < \tau < 1$. The discrepancy stems from the formula $2/(1+\tau)$ being the classical Jarník exponent for the normalization $q^\tau \|q\alpha\|$, whereas the conjecture’s definition uses $q^{1/\tau}$ — the two parametrizations are confused.

4 Verification

The arithmetic core of the attack is machine-checked in core Lean 4 (no Mathlib), in the `lean4/` directory: `Tlmc325.attack` proves `gtFrac 4 3 1 1 = true` (i.e. $4/3 > 1$), `Tlmc325.claimed_exceeds_ambient` proves the general $2q/(p+q) > 1$ for $q > p$, and two mismatch theorems record the true values at $\tau = 2$ and $\tau = 1/4$. Every declaration is proved with zero axioms (`Check.lean` prints *does not depend on any axioms* for all of them) and no `sorry`. The script `reproduce.py` recomputes all numerics standalone with exact rational arithmetic.